Functional Requirements Document – Project Management

TaskFlow – Intelligent Task Allocation & Tracking System

> **Document:** Functional Requirements Document (sub-module PROJ) · **Version:** 1.0 · **Status:** Draft **Project type:** Personal Project (Skill Development) · **Date:** 31-Oct-2026 **Author:** Jojo George (jojogeorge3344@gmail.com) **Module:** TaskFlow Management · **Sub-module:** 03.2 Project Management, including Positions (code `PROJ`) **Based on:** `BRD-Project-Management.md` v1.0 © 2026 Jojo George. All rights reserved.

Document Control

Revision History

Version	Date	Author	Summary of changes
1.0	31-Oct-2026	Jojo George	Initial version, written from the Project Management BRD v1.0 and TaskFlow as built

Related Documents

Document	Purpose	Location
BRD – Project Management	Business needs this FRD details (`REQ-PROJ-nn`, `BR-PROJ-nn`)	`../../../BRD/03-TaskFlow-Management/03.2-Project-Management/BRD-Project-Management.md`
BRD – Overall	Roles, global rules `BR-nn`, global NFRs	`../../../BRD/00-Overall/BRD-TaskFlow-Overall.md`
FRD – Overall	Shared roles, cross-cutting behaviour, index	`../../00-Overall/FRD-TaskFlow-Overall.md`
Related FRDs	Employee Management, Department Management, Task Management	`../03.1-Employee-Management/FRD-Employee-Management.md`, `../03.3-Department-Management/FRD-Department-Management.md`, `../../02-TaskFlow-Monitoring/02.1-Task-Management/FRD-Task-Management.md`
Development Plan	No Project Management screen in the plan (projects were added during development)	`TF - Documents/Screens Wise Feature Development Plan/`
HLD / LLD	Technical design	`../../../HLD/03-TaskFlow-Management/03.2-Project-Management/HLD-Project-Management.md`, `../../../LLD/03-TaskFlow-Management/03.2-Project-Management/LLD-Project-Management.md`
TDD	Technical design document (whole system)	`TF - Documents/TF - TDD/TDD-TaskFlow.md`

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1. Introduction

1.1 Purpose

This FRD specifies the detailed, testable behaviour of the Project Management sub-module: the project list, the add / edit project dialog, the project team dialog (including positions) and the Project Details screen – their inputs, outputs, rules, validations, messages and permissions.

1.2 Scope

In scope: listing and searching projects; adding, editing and deleting projects; building each project's team with position, allocation percentage and project-manager flag; picking or creating positions; Project Details with summary figures, team and a filtered task list.

Out of scope (BRD section 3.2): renaming or deleting positions; project dates, budgets or status; creating tasks from the project screens (Task Management); changing allocations from Employee Management after creation.

1.3 Audience

Developer, tester, reviewer and product owner of TaskFlow (all Jojo George).

1.4 Conventions

- Requirement IDs: `FR-PROJ-nn`. Use cases: `UC-PROJ-nn`. Business rules: `BR-PROJ-nn` (Project Management BRD) and `BR-nn` (overall BRD). BRD requirements: `REQ-PROJ-nn`.
- Priority: **Must / Should / Could** (MoSCoW), taken from the BRD requirement traced.
- "The system shall …" = mandatory behaviour.
- `{name}`, `{n}`, `{total}`, `{other}`, `{left}`, `{names}` in messages are replaced by the real values.
- Permission "System" = a rule the system applies whatever the user's permissions.

1.5 Definitions & Abbreviations

Term	Meaning
Project	A body of work that tasks belong to
Live project	A project that has not been deleted
Team	The people allocated to a project
Allocation	The whole percentage (1–100) of a person's time given to one project
Project manager	A team member with the Manager role, flagged to manage the project
Position	A shared label for a person's job on a project (for example Developer); it grants no access
Manager's team	People in the departments a Manager manages plus people on the projects they manage
Soft delete	Hiding a record from every list while keeping it and its history
Scope	Own / Team / Organisation data visibility; Project Management is Organisation

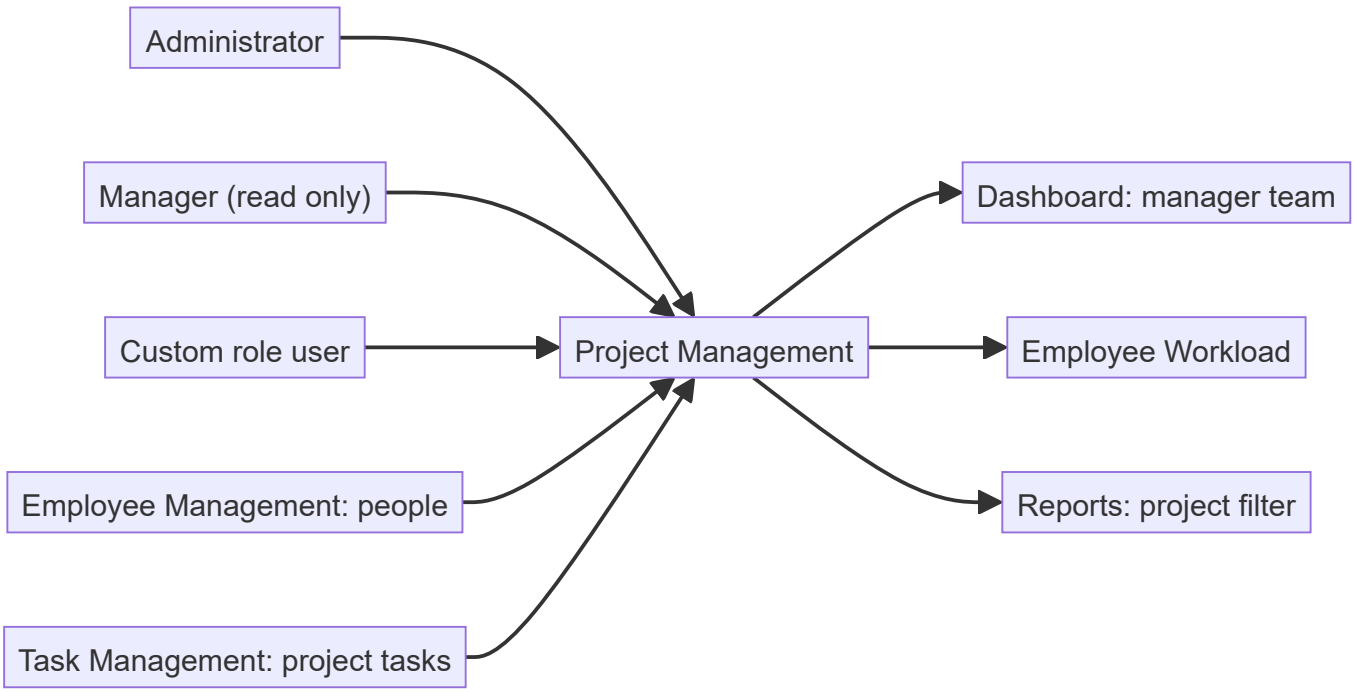
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2. System Overview

2.1 Module Map

Code	Module	Child modules / screens	Primary roles
PROJ	TaskFlow Management → Project Management (this FRD)	Project Management list, Add / Edit project (dialog), Project team / Assign team (dialog) with Positions, Project Details	Administrator (full), Manager (read only)
EMP	Employee Management (related)	Supplies people for the team picker; sets first allocations when an employee is added	Administrator
DEPT	Department Management (related)	Combined with project managers to form a manager's team	Administrator
TASK	Task Management (related)	Supplies the project's tasks; every task belongs to a project	Manager, Employee
DASH / WORK / RPT	Dashboard, Employee Workload, Reports (related)	Read projects, teams and allocations	All roles as permitted

2.2 Context Diagram



3. User Roles & Permissions

3.1 Roles

Role	Description	Data scope
Administrator	Adds, edits and deletes projects, manages teams, creates positions	Organisation
Manager	Views projects, Project Details and positions read only; can be flagged as a project manager	Organisation, read only
Employee	No access to this sub-module	None

Role	Description	Data scope
Custom role	Needs `projects.view` to see the screens; can do what it is granted	As granted (Organisation)

3.2 Permission Matrix (functional view)

Module › Function	Permission key	Administrator	Manager	Employee	Custom role
Projects › View list, Project Details, team, tasks, positions	`projects.view`	Granted	Granted	Denied	Configurable
Projects › Add project	`projects.create`	Granted	Denied	Denied	Configurable
Projects › Edit project	`projects.edit`	Granted	Denied	Denied	Configurable
Projects › Delete project	`projects.delete`	Granted	Denied	Denied	Configurable
Projects › Manage project team	`projects.members.manage`	Granted	Denied	Denied	Configurable
Projects › Create position	`positions.create`	Granted	Denied	Denied	Configurable
Employees › People picker in the team dialog (supporting)	`employees.view`	Granted	Granted	Denied	Configurable

Every key except `projects.view` also requires `projects.view`. Older roles may still hold `projects.manage`, which acts like the finer project keys it was split into.

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4. Functional Requirements by Module

4.1 Module: Project Management (PROJ)

Purpose: Keep one list of projects, staff each one without over-booking anyone, and give every team a responsible manager. · **BRD reference:** BRD-Project-Management SCREEN 1–4, BR-PROJ-01 – BR-PROJ-11; overall BR-06, BR-21

Change from original BRD: the original BRD has no Project Management screen; projects, teams, allocations, positions and Project Details are new requirements recorded here as built.

4.1.1 Screen: Project Management (list)

Field	Value
Route / location	`/projects` – menu **TaskFlow Management → Project Management**
Access	Administrator (full); Manager (read only); custom roles with `projects.view`. Employee: no access (route redirects to the dashboard)
Entry points	Side menu

Screen elements

Element	Type	Description	Mandatory	Visible to
Project count	Header text	Number of projects	–	Users with `projects.view`
Add project	Button	Opens the Add project dialog	–	Users with `projects.create`
Search by name or description	Text input	Narrows the cards	No	Users with `projects.view`
Project card	Card	Name (link to Project Details), description, "{n} employees", **Manager/Managers** names	–	Users with `projects.view`
View employees	Link	Opens Employee Management filtered by the project	–	Users who can open Employee Management
Team	Card button	Opens the Project team dialog	–	Users with `projects.members.manage`
Edit	Card button	Opens the Edit project dialog	–	Users with `projects.edit`
Delete	Card button	Deletes the project after confirmation	–	Users with `projects.delete`

Functional requirements

ID	Requirement	Priority	Permission	BRD trace
FR-PROJ-01	The system shall show every live project as a card with its name, description, "{n} employees" and the names of its project managers under **Manager** or **Managers**.	Must	`projects.view`	REQ-PROJ-01
FR-PROJ-02	The system shall show the number of projects in the header.	Should	`projects.view`	REQ-PROJ-02
FR-PROJ-03	The system shall narrow the cards to projects whose name or description contains the search text.	Should	`projects.view`	REQ-PROJ-03
FR-PROJ-04	The system shall show the Project Management menu item and open `/projects` and `/projects/{id}` only for users holding `projects.view`, and redirect other users to their dashboard.	Must	`projects.view`	REQ-PROJ-04
FR-PROJ-05	The system shall show **Add project**, **Team**, **Edit** and **Delete** only to users holding `projects.create`, `projects.members.manage`, `projects.edit` and `projects.delete` respectively, so a Manager sees the list read only.	Must	As listed	REQ-PROJ-05
FR-PROJ-06	The system shall refuse every project action requested without the matching permission, even when requested outside the screen, and show the action's failure message.	Must	As listed	REQ-PROJ-05, NFR-PROJ-03, BR-04
FR-PROJ-07	The system shall show **View employees** only to users who can open Employee Management, and open the employee list filtered by that project.	Could	Any employee action permission	REQ-PROJ-06

ID	Requirement	Priority	Permission	BRD trace
FR-PROJ-08	The system shall ask "Delete project?" – ""{name}" will be permanently deleted and its {n} employees will be taken off it." before deleting, using only the first sentence part when the project has no team.	Must	`projects.delete`	REQ-PROJ-07
FR-PROJ-09	The system shall, on delete, remove every allocation and project-manager link of the project so each member's percentage is free again, and hide the project by soft delete.	Must	`projects.delete`	REQ-PROJ-08, BR-PROJ-09, BR-21
FR-PROJ-10	The system shall keep the project's tasks when the project is deleted.	Must	System	REQ-PROJ-08, BR-PROJ-09
FR-PROJ-11	The system shall confirm a deletion with "{name} deleted", or show "Could not delete the project." when refused.	Should	`projects.delete`	REQ-PROJ-08
FR-PROJ-12	The system shall show "No projects yet – Add the first project to start organising employees." when there are no projects, and "Could not load projects. Check your connection and try again." when loading fails.	Could	`projects.view`	REQ-PROJ-09
FR-PROJ-13	The system shall show projects to every holder of `projects.view` organisation-wide, without a team limit.	Must	`projects.view`	BR-PROJ-11

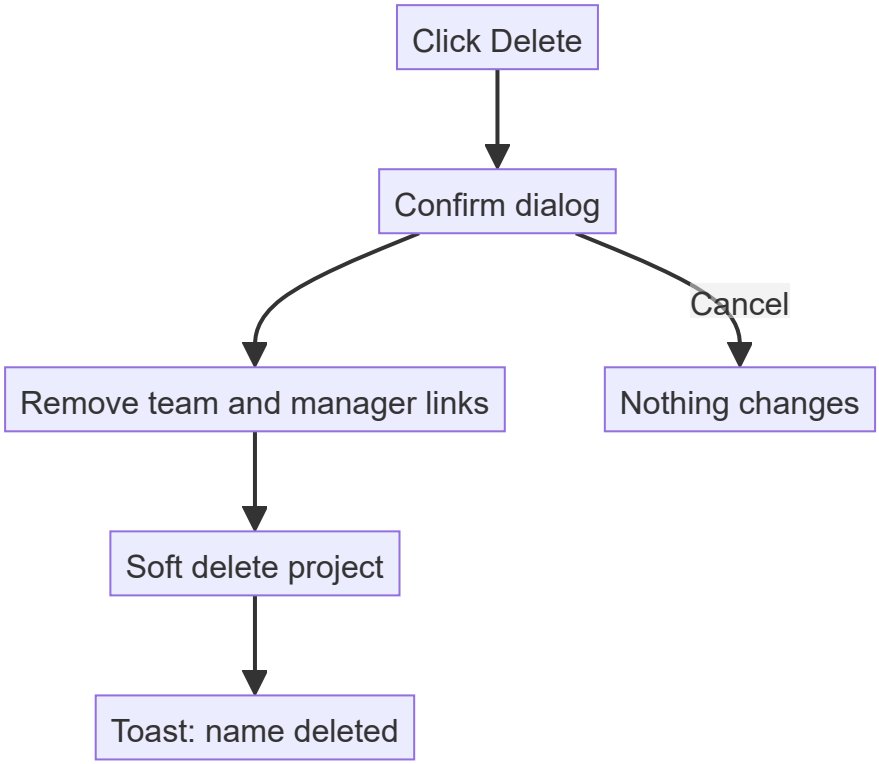
Use case: UC-PROJ-01 – Browse projects

Item	Detail
Actor	Manager (or Administrator)
Preconditions	Signed in with MFA; holds `projects.view`
Trigger	Opens **TaskFlow Management → Project Management**
Main flow	1. System shows the project count and one card per live project. 2. User types in the search box. 3. System narrows the cards. 4. User clicks a project name → Project Details (UC-PROJ-07).
Alternate flows	A1: user can open Employee Management → **View employees** opens the employee list filtered by the project.
Exceptions	E1: no projects → empty-state message. E2: load fails → "Could not load projects. Check your connection and try again."
Postconditions	None (read only).

Use case: UC-PROJ-02 – Delete a project

Item	Detail
Actor	Administrator
Preconditions	Holds `projects.delete`
Trigger	Clicks **Delete** on a card
Main flow	1. System asks "Delete project?" with the member count. 2. Administrator confirms. 3. System removes the team and manager links and soft-deletes the project. 4. System shows "{name} deleted".
Alternate flows	–

Item	Detail
Exceptions	E1: refused → "Could not delete the project."
Postconditions	Project hidden; members' allocation freed; project tasks kept.



Field validations: *Not applicable for this screen* (no input other than search).

Business rules applied: BR-PROJ-09, BR-PROJ-11; BR-21

Notifications / side effects

Event	Who is notified	Channel
Project deleted	Nobody; members' allocations are freed	– (no notification, no email)

Messages

Situation	Message
Deleted	"{name} deleted"
Delete refused	"Could not delete the project."
No permission	Buttons hidden; `/projects` redirects to the dashboard without `projects.view`

4.1.2 Screen: Add / Edit Project (dialog)

Field	Value
Route / location	Dialog on `/projects` – **Add project** / **Edit project**
Access	Administrator; custom roles with `projects.create` (add) or `projects.edit` (edit). Manager and Employee: no access
Entry points	**Add project** button; **Edit** on a card

Screen elements

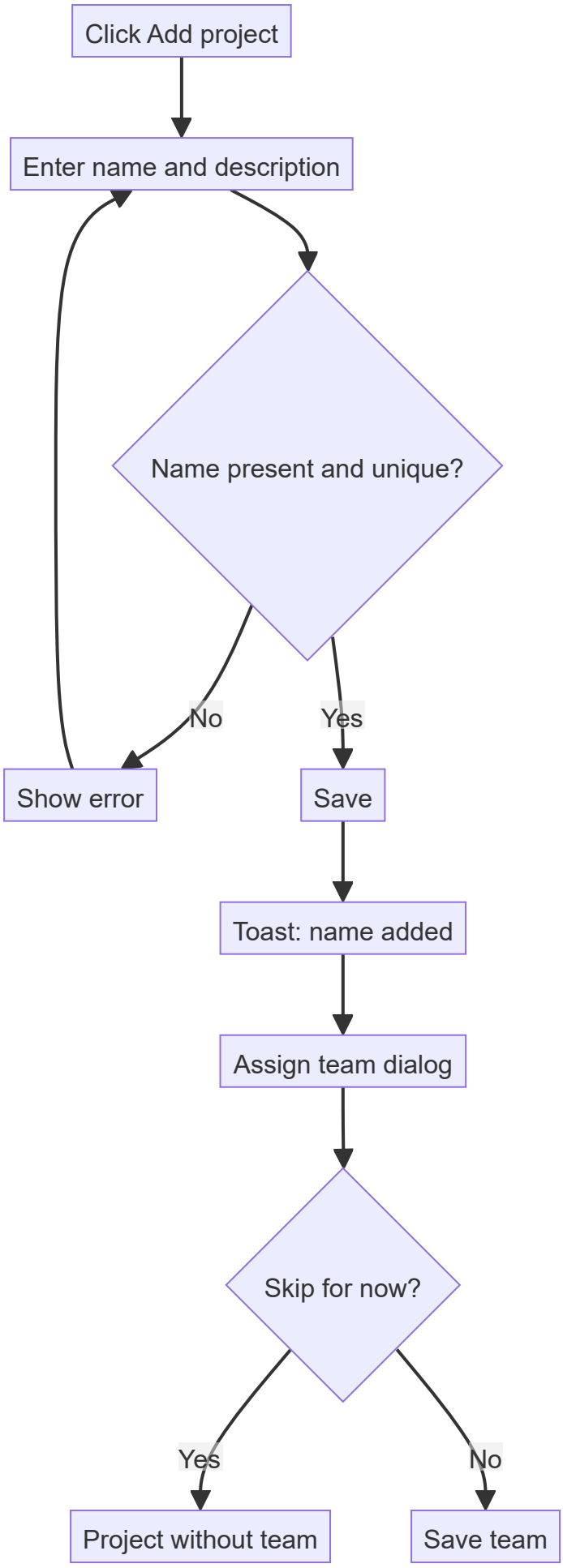
Element	Type	Description	Mandatory	Visible to
Name	Text input	Project name	Yes	Users with `projects.create` / `projects.edit`
Description	Text area	Optional description	No	Same
Add project / Save	Button	Submits the dialog	–	Same

Functional requirements

ID	Requirement	Priority	Permission	BRD trace
FR-PROJ-14	The system shall create a project only with a name of at most 200 characters and an optional description of at most 500 characters.	Must	`projects.create`	REQ-PROJ-10, BR-PROJ-02
FR-PROJ-15	The system shall reject a name already used by a live project, ignoring case, with "Project name already exists."	Must	System	REQ-PROJ-11, BR-PROJ-02
FR-PROJ-16	The system shall let the user change a project's name and description under the same rules.	Must	`projects.edit`	REQ-PROJ-12
FR-PROJ-17	The system shall, straight after a project is created, open the team dialog titled **Assign team: {name}** with a **Skip for now** button.	Should	`projects.create`	REQ-PROJ-13
FR-PROJ-18	The system shall confirm a save with "{name} added" or "{name} updated", or show "Could not add the project." / "Could not update the project." when refused.	Should	`projects.create`, `projects.edit`	REQ-PROJ-14

Use case: UC-PROJ-03 – Add a project

Item	Detail
Actor	Administrator
Preconditions	Holds `projects.create`
Trigger	Clicks **Add project**
Main flow	1. Administrator enters Name and optional Description. 2. Clicks **Add project**. 3. System checks the name is unique and saves. 4. System shows "{name} added". 5. System opens **Assign team: {name}** (UC-PROJ-05).
Alternate flows	A1: **Skip for now** → project saved with no team; the team can be added later with **Team**.
Exceptions	E1: name missing → "Name is required." E2: duplicate → "Project name already exists."
Postconditions	Project listed; team set or left empty.


Click Add project
Enter name and description
Name present and unique?
No
Yes
Show error
Save
Toast: name added
Assign team dialog
Skip for now?
Yes
No
Project without team
Save team

Use case: UC-PROJ-04 – Edit a project

Item	Detail
Actor	Administrator
Preconditions	Holds `projects.edit`
Trigger	Clicks **Edit** on a card
Main flow	1. System opens Edit project with the saved values. 2. Administrator changes Name or Description and clicks **Save**. 3. System validates and saves. 4. System shows "{name} updated".
Alternate flows	–
Exceptions	E1: name missing or duplicate → field message. E2: refused → "Could not update the project."
Postconditions	Project renamed or re-described everywhere it is shown.

Field validations

Field	Rule	Error message
Name	Required; at most 200 characters; unique among live projects, ignoring case	"Name is required." · "Project name already exists."
Description	Optional; at most 500 characters	⚠️ To be confirmed: exact over-length message not recorded

Business rules applied: BR-PROJ-02

Notifications / side effects

Event	Who is notified	Channel
Project added or edited	Nobody	–

Messages

Situation	Message
Success	"{name} added" · "{name} updated"
Refused	"Could not add the project." / "Could not update the project."

4.1.3 Screen: Project Team / Assign Team (dialog), including Positions

Field	Value
Route / location	Dialog on `/projects` – **Project team**, titled **Assign team: {name}** when opened right after creation
Access	Administrator; custom roles with `projects.members.manage` (people picker also needs `employees.view`); creating a position needs `positions.create`. Manager and Employee: no access
Entry points	**Team** on a card; automatically after **Add project**; the Position field is also used in the Add employee panel (Employee Management)

Screen elements

Element	Type	Description	Mandatory	Visible to
Employee	Picker per row	Active person; shows "{n}% remaining" or "Fully allocated"	Yes	Users with `projects.members.manage`
Position	Pick-or-type field per row	Existing position or a new name; new rows start with **Member**	Yes	Same (creating needs `positions.create`)
Allocation %	Number input per row	Whole percentage 1–100	Yes	Same
Project manager	Tick box per row	Enabled only for people with the Manager role; tooltip "Only a person with the Manager role can manage a project"	–	Same
Remove	Row button	Removes the row	–	Same
Add employee	Button	Adds a row	–	Same
Save team	Button	Saves the whole team	–	Same
Skip for now	Button	Closes without a team (after creation only)	–	Same

Functional requirements

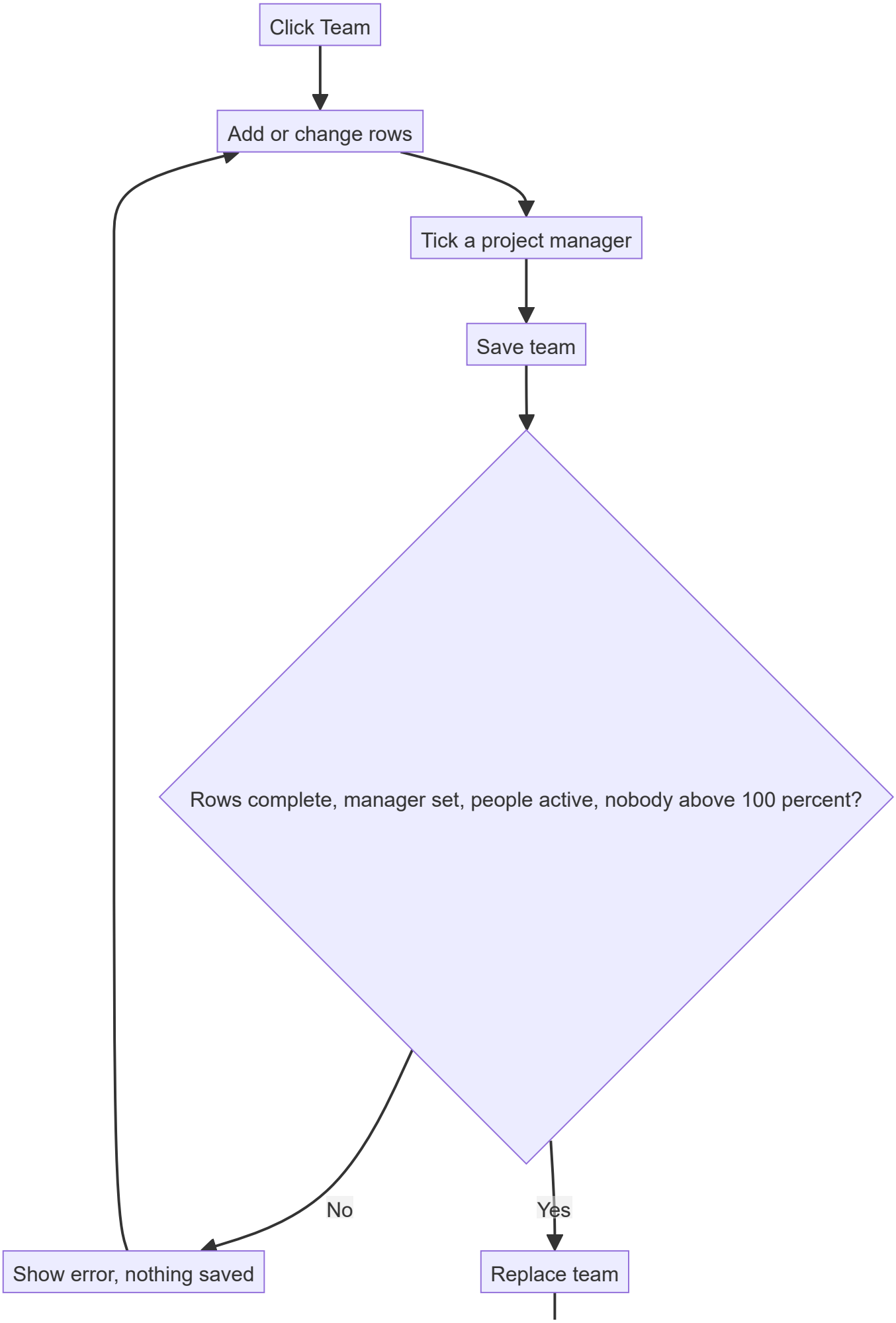
ID	Requirement	Priority	Permission	BRD trace
FR-PROJ-19	The system shall load the current team, one row per person with position, allocation percentage and project-manager flag, or show "Could not load the team. Close this and try again."	Must	`projects.members.manage`	REQ-PROJ-15
FR-PROJ-20	The system shall show, when picking a person, "{n}% remaining" or "Fully allocated" based on their allocations on other projects.	Should	`projects.members.manage`, `employees.view`	REQ-PROJ-16, NFR-PROJ-04
FR-PROJ-21	The system shall flag an over-capacity row before saving with "{name} is fully allocated on other projects." or "{name} has only {n}% remaining."	Should	System	REQ-PROJ-16, REQ-PROJ-22, NFR-PROJ-04
FR-PROJ-22	The system shall require every row to have an employee, a position and a whole percentage from 1 to 100.	Must	System	REQ-PROJ-17, BR-PROJ-04
FR-PROJ-23	The system shall allow a person only once on a project.	Must	System	REQ-PROJ-18, BR-PROJ-04
FR-PROJ-24	The system shall refuse a non-empty team in which no one is ticked as project manager.	Must	System	REQ-PROJ-19, BR-PROJ-05
FR-PROJ-25	The system shall allow only people with the Manager role to be ticked as project manager, and refuse others with "Only a person with the Manager role can manage a project: {names}."	Must	System	REQ-PROJ-20, BR-PROJ-05
FR-PROJ-26	The system shall accept only existing, active people on a team.	Must	System	REQ-PROJ-21, BR-PROJ-06

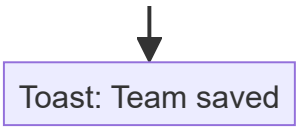
ID	Requirement	Priority	Permission	BRD trace
FR-PROJ-27	The system shall refuse a team in which any person's total across all projects would pass 100%, with "{name} would be allocated {total}% in total. They already have {other}% on other projects, so {left}% remains."	Must	System	REQ-PROJ-22, BR-PROJ-03
FR-PROJ-28	The system shall save the whole team or nothing: if any rule fails, no row changes, and the 100% cap holds even when two administrators save teams at the same time.	Must	System	REQ-PROJ-23, BR-PROJ-07, NFR-PROJ-01, NFR-PROJ-02
FR-PROJ-29	The system shall replace the project's team with the saved rows and confirm with "Team saved", or "Team cleared" when no one is left; a refused save shows "Could not save the team."	Should	`projects.members.manage`	REQ-PROJ-24
FR-PROJ-30	The system shall keep the project's manager links in step with the project-manager flags, so the flagged Managers' teams include everyone on the project for the manager dashboard and task visibility.	Must	System	REQ-PROJ-19, BR-PROJ-10
FR-PROJ-31	The system shall start each new row with the position **Member** when that position exists.	Could	System	REQ-PROJ-25
FR-PROJ-32	The system shall let the user pick a position or type a name, and reuse an existing position whose name matches ignoring case.	Must	`projects.view`	REQ-PROJ-26, BR-PROJ-08
FR-PROJ-33	The system shall create a typed position that does not exist and make it available in every position list; a failure shows "The position could not be created."	Must	`positions.create`	REQ-PROJ-27, BR-PROJ-08
FR-PROJ-34	The system shall require a position name, remove extra spaces and reject names longer than 100 characters.	Must	System	REQ-PROJ-28, BR-PROJ-08
FR-PROJ-35	The system shall provide the built-in positions Member, Manager, QA, Developer, DevOps and System Admin.	Should	System	REQ-PROJ-29
FR-PROJ-36	The system shall treat positions as labels only: holding a position grants no permission, and positions cannot be renamed or deleted.	Must	System	REQ-PROJ-30, BR-PROJ-08

Use case: UC-PROJ-05 – Manage a project team

Item	Detail
Actor	Administrator
Preconditions	Holds `projects.members.manage` and `employees.view`
Trigger	Clicks **Team** on a card, or the dialog opens after **Add project**
Main flow	1. System loads the team and positions. 2. Administrator clicks **Add employee** and picks a person (picker shows remaining percentage). 3. Picks or types a Position and enters Allocation %. 4. Ticks **Project manager** for at least one Manager. 5. Clicks **Save team**. 6. System checks every rule for the whole team and replaces it. 7. System shows "Team saved".

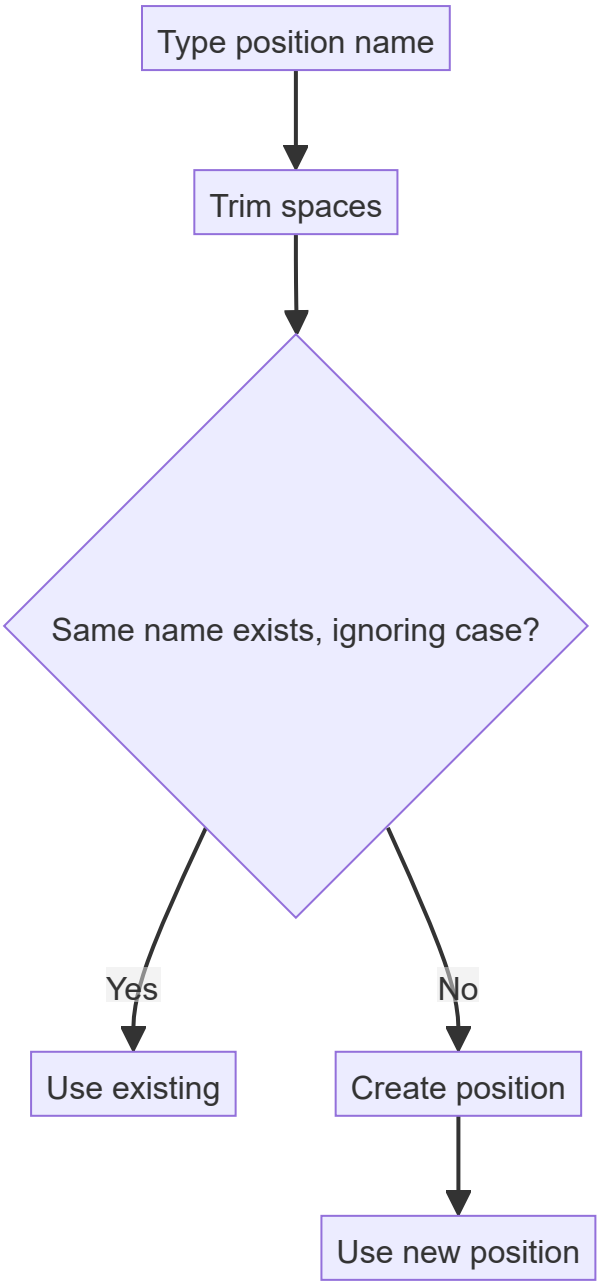
Item	Detail
Alternate flows	A1: all rows removed → "Team cleared". A2: typed position is new → created and used (UC-PROJ-06). A3: opened after creation → **Skip for now** closes without a team.
Exceptions	E1: incomplete row → "Give every row an employee, a position and a percentage, or remove the row." E2: no manager → "Mark at least one person as the project manager." E3: non-Manager ticked → "Only a person with the Manager role can manage a project: {names}." E4: inactive person → "One or more people do not exist or are not active." E5: over 100% → capacity message (FR-PROJ-27). In every exception nothing is saved.
Post conditions	Team replaced; managers' teams updated; nobody above 100%.


Click Team
Add or change rows
Tick a project manager
Save team
Rows complete, manager set, people active, nobody above 100 percent?
No
Yes
Show error, nothing saved
Replace team



Use case: UC-PROJ-06 – Create a position

Item	Detail
Actor	Administrator
Preconditions	Holds `positions.create`; a Position field is open (team dialog or Add employee panel)
Trigger	Types a position name and commits the field
Main flow	1. System trims extra spaces. 2. System looks for a position with the same name, ignoring case. 3. If none exists, system creates it. 4. The row uses the position.
Alternate flows	A1: "qa" typed → existing "QA" is used, nothing created.
Exceptions	E1: empty → "Position name is required." E2: too long → "Position name must be 100 characters or fewer." E3: creation fails or not permitted → "The position could not be created."
Postconditions	Position available in every position list.



Field validations

Field	Rule	Error message
Row	Employee, position and percentage all set	"Give every row an employee, a position and a percentage, or remove the row."
Allocation %	Whole number 1–100	"Use a whole percentage from 1 to 100." (form) · "Allocation percentage must be a whole number from 1 to 100." (server)
Employee	Once per project; exists and active	"A person can only appear once on a project." · "One or more people do not exist or are not active."
Employee total	At most 100% across all projects	"{name} is fully allocated on other projects." · "{name} has only {n}% remaining." (form) · "{name} would be allocated {total}% in total. They already have {other}% on other projects, so {left}% remains." (server)
Project manager	At least one on a non-empty team; Manager role only	"Mark at least one person as the project manager." (form) / "Mark at least one person on the team as the project's manager." (server) · "Only a person with the Manager role can manage a project: {names}."

Field	Rule	Error message
Position name	Required; at most 100 characters; trimmed; unique ignoring case	"Position name is required." · "Position name must be 100 characters or fewer." · "The position could not be created."

Business rules applied: BR-PROJ-03 – BR-PROJ-08, BR-PROJ-10

Notifications / side effects

Event	Who is notified	Channel
Team saved	Nobody; flagged Managers' teams change	– (no notification, no email)

Messages

Situation	Message
Success	"Team saved" · "Team cleared"
Load / save errors	"Could not load the team. Close this and try again." · "Could not save the team."
No permission	**Team** hidden; a refused call shows "Could not save the team."

4.1.4 Screen: Project Details

Field	Value
Route / location	`/projects/{id}`
Access	Administrator, Manager; custom roles with `projects.view`. Employee: no access
Entry points	Project name on a card

Screen elements

Element	Type	Description	Mandatory	Visible to
Project name	Heading	Name of the project	–	Users with `projects.view`
Summary figures	Tiles	Members, Tasks, **Overdue**, **Completed** (percentage)	–	Same
Team	List	Person, position, percentage	–	Same
Tasks	Table	Task, Assignee, Priority, Status, Deadline, "Overdue" badge	–	Same
Status, Priority, Assigned to, Role, Deadline from, Deadline to	Filters	Narrow the task table; Role = position on this project	No	Same
Clear filters	Button	Resets all task filters	–	Same

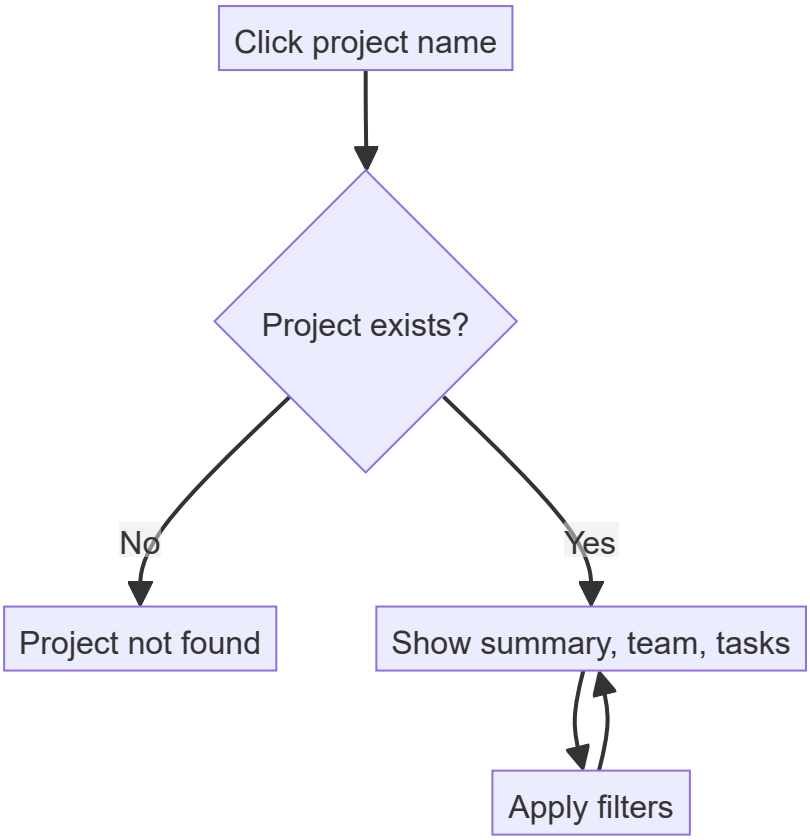
Functional requirements

ID	Requirement	Priority	Permission	BRD trace
FR-PROJ-37	The system shall open Project Details when the user clicks a project name.	Must	`projects.view`	REQ-PROJ-31

ID	Requirement	Priority	Permission	BRD trace
FR-PROJ-38	The system shall show the project's number of members, number of tasks, number of overdue tasks and completed percentage.	Should	`projects.view`	REQ-PROJ-32
FR-PROJ-39	The system shall list the team with each person's position and percentage, or "No one is assigned to this project yet."	Must	`projects.view`	REQ-PROJ-33
FR-PROJ-40	The system shall list all of the project's tasks with task, assignee, priority, status and deadline, soonest deadline first, and mark overdue tasks with an "Overdue" badge.	Must	`projects.view`	REQ-PROJ-34, BR-12
FR-PROJ-41	The system shall filter the tasks by status, priority, assignee, the assignee's position on this project and a deadline range, applying all set filters together.	Should	`projects.view`	REQ-PROJ-35
FR-PROJ-42	The system shall reset every task filter when the user clicks **Clear filters**.	Could	`projects.view`	REQ-PROJ-36
FR-PROJ-43	The system shall show "Project not found – It may have been deleted, or the link is wrong." for a deleted or unknown project.	Must	`projects.view`	REQ-PROJ-37
FR-PROJ-44	The system shall show "No tasks yet – Tasks created for this project will appear here." when the project has no tasks, and "Could not load the project.", "Could not load the project tasks." or "Could not load the tasks." when loading fails.	Should	`projects.view`	REQ-PROJ-33, REQ-PROJ-34
FR-PROJ-45	The system shall show Project Details, its team and its tasks to every holder of `projects.view`, organisation-wide.	Must	`projects.view`	BR-PROJ-11, BR-PROJ-01

Use case: UC-PROJ-07 – View project details

Item	Detail
Actor	Manager (or Administrator)
Preconditions	Holds `projects.view`
Trigger	Clicks a project name
Main flow	1. System shows the summary figures, the team and all tasks, soonest deadline first. 2. User sets Status, Priority, Assigned to, Role or deadline filters. 3. System reloads the tasks with all filters applied. 4. User clicks **Clear filters** to see all tasks again.
Alternate flows	–
Exceptions	E1: project deleted or unknown → "Project not found – It may have been deleted, or the link is wrong." E2: load fails → load-error message.
Postconditions	None (read only).



Field validations

Field	Rule	Error message
Deadline from / to	Date filters; both optional	⚠️ To be confirmed: no message is recorded for a "from" date after the "to" date

Business rules applied: BR-PROJ-01, BR-PROJ-11; BR-12, BR-22

Notifications / side effects: *Not applicable for this screen* (read only).

Messages

Situation	Message
Not found	"Project not found – It may have been deleted, or the link is wrong."
Empty	"No one is assigned to this project yet." · "No tasks yet – Tasks created for this project will appear here."
Load errors	"Could not load the project." · "Could not load the project tasks." · "Could not load the tasks."

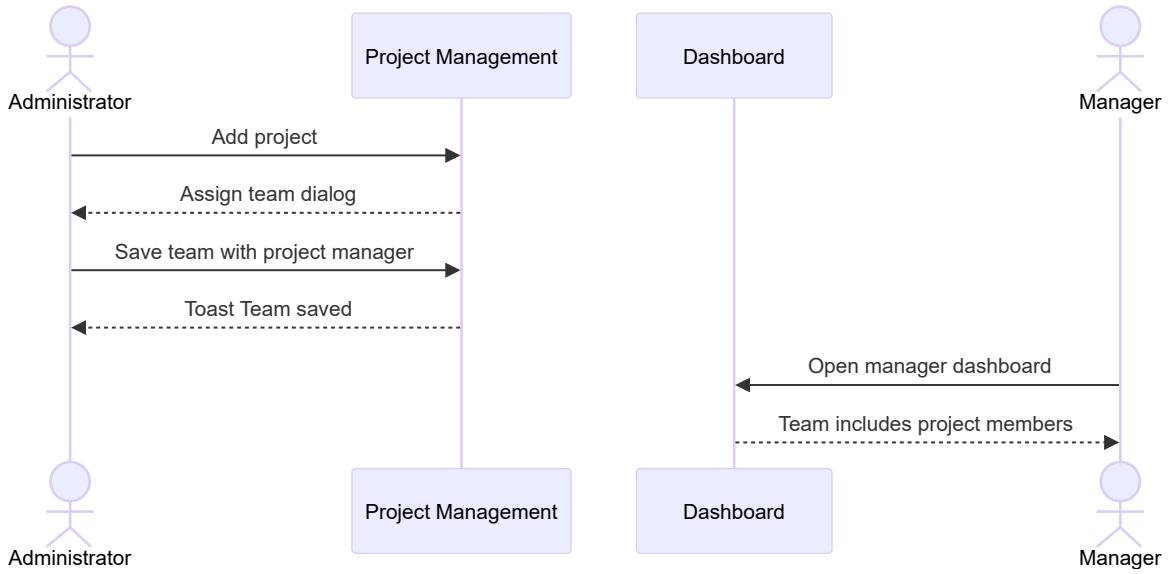
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5. Cross-Module Workflows

5.1 New project to manager dashboard

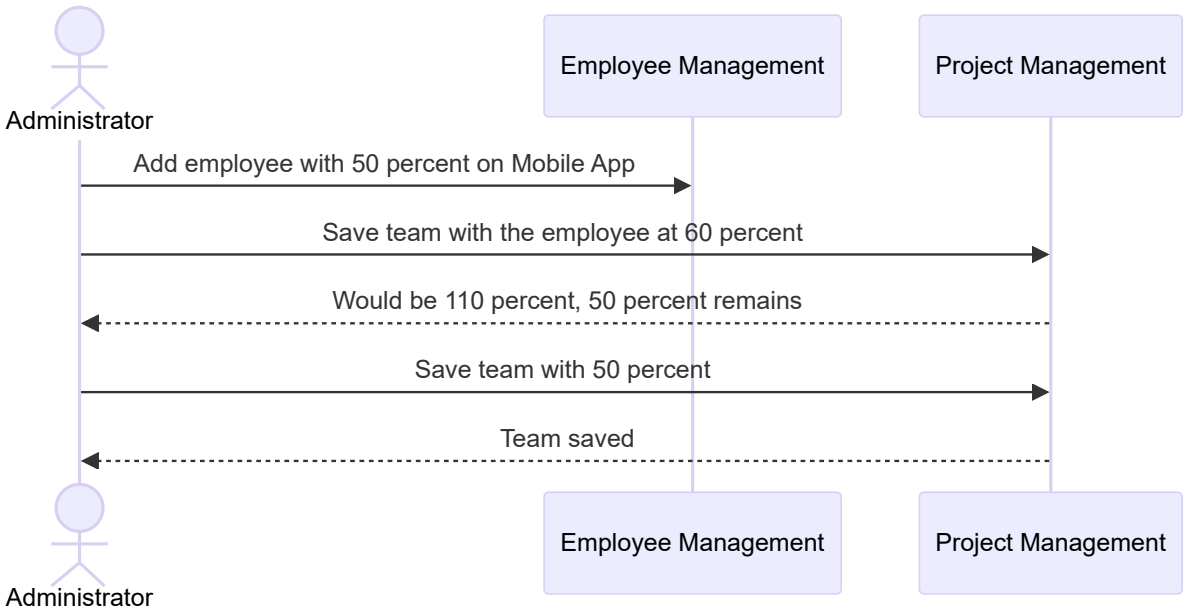
Step	Module	Actor / System	Action	FR reference
1	PROJ	Administrator	Adds project *Website Revamp*	FR-PROJ-14, FR-PROJ-17
2	PROJ	Administrator	Adds Ravi Menon (Manager, 30%, project manager) and Anu Thomas (Developer, 60%) and saves	FR-PROJ-24 – FR-PROJ-29
3	PROJ	System	Makes Anu part of Ravi's team	FR-PROJ-30

Step	Module	Actor / System	Action	FR reference
4	DASH / TASK	System	Ravi's manager dashboard and task visibility include Anu	Dashboard FRD, Task Management FRD



5.2 Shared 100% cap with Employee Management

Step	Module	Actor / System	Action	FR reference
1	EMP	Administrator	Adds Anu Thomas with 50% on *Mobile App*	FR-EMP-49, FR-EMP-52
2	PROJ	Administrator	Adds Anu to *Website Revamp* at 60%	FR-PROJ-20, FR-PROJ-21
3	PROJ	System	Refuses: 110% in total, 50% remains	FR-PROJ-27, FR-PROJ-28
4	PROJ	Administrator	Changes the row to 50% and saves	FR-PROJ-29



5.3 Tasks appear in Project Details

Step	Module	Actor / System	Action	FR reference
1	TASK	Manager	Creates a task for *Website Revamp*	Task Management FRD

Step	Module	Actor / System	Action	FR reference
2	PROJ	Manager	Opens Project Details and filters by assignee	FR-PROJ-40, FR-PROJ-41
3	PROJ	Administrator	Deletes the project later; tasks stay	FR-PROJ-09, FR-PROJ-10

6. Reports & Exports

Not applicable for this sub-module. Project Details shows per-project summary figures (FR-PROJ-38); formal reports filter by project in Reports (see the Reports FRD).

7. Notifications & Messages Catalogue

Project actions send no notification and no email. Success messages:

Code	Trigger	Recipients	Channel	Text
M-PROJ-01	Project added	Acting user	Toast	"{name} added"
M-PROJ-02	Project updated	Acting user	Toast	"{name} updated"
M-PROJ-03	Project deleted	Acting user	Toast	"{name} deleted"
M-PROJ-04	Team saved	Acting user	Toast	"Team saved" / "Team cleared"

8. Data Requirements (functional view)

Data item	Description	Created by	Rules
Project	Name and description	Administrator	Name required, at most 200, unique among live projects ignoring case; description at most 500; soft delete
Team row (allocation)	Employee, project, position, percentage, project-manager flag	Administrator (team dialog or Add employee)	Whole 1–100; person once per project; total per person at most 100%; removed when the project is deleted
Project manager link	Which Managers manage the project	System (from the flags)	Manager role only; removed on project delete or when the person stops being a Manager
Position	Shared label	System (seed) or Administrator (typed)	Required, at most 100, trimmed, unique ignoring case; seeded Member, Manager, QA, Developer, DevOps, System Admin; no rename or delete

9. Non-Functional Requirements (functional impact)

The global NFRs of the overall BRD apply (notably NFR-08 server validation, NFR-12 usability).

ID	Requirement	Affects
NFR-PROJ-01	No person's stored total ever exceeds 100%, even with simultaneous team saves	FR-PROJ-27, FR-PROJ-28
NFR-PROJ-02	A team save changes every row or none	FR-PROJ-28
NFR-PROJ-03	Permission checks are enforced by the server	FR-PROJ-06
NFR-PROJ-04	Allocation feedback is shown before saving	FR-PROJ-20, FR-PROJ-21

ID	Requirement	Affects
NFR-PROJ-05	Every refusal states the reason in plain words	Messages in section 4

10. Assumptions & Open Issues

Type	Description	Owner
Assumption	"100%" means a person's full working time; part-time work is not modelled separately.	Jojo George
Assumption	Only people with the Manager system role manage projects; custom roles cannot be project managers.	Jojo George
Open issue	⚠️ To be confirmed: Project Details shows every task of the project to anyone with `projects.view`, regardless of their task-view scope (R-PROJ-02).	Jojo George
Open issue	⚠️ To be confirmed: how tasks of a deleted project should be shown (R-PROJ-03).	Jojo George
Open issue	⚠️ To be confirmed: the Assign team dialog opens after creation and the Position field tries to create positions even for users without `projects.members.manage` / `positions.create`; the server refuses the save (R-PROJ-04).	Jojo George
Open issue	⚠️ To be confirmed: the delete confirmation says "permanently deleted" although deletion is a soft delete (R-PROJ-05).	Jojo George
Open issue	Positions cannot be renamed or deleted, so misspelt positions stay (R-PROJ-06); mitigated by FR-PROJ-32 and FR-PROJ-34.	Jojo George

11. Requirements Traceability Matrix

Test evidence lists the automated test suites that cover the behaviour; the mapping was checked against the test sources. Frontend specs (`*.spec.ts`) are under `frontend/src/app`. "Manual UI" = screen behaviour checked by hand.

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-PROJ-01	FR-PROJ-01	4.1.1	UC-PROJ-01	UserAllocationServiceTests (managers by name), project-list.component.spec.ts
REQ-PROJ-02	FR-PROJ-02	4.1.1	UC-PROJ-01	project-list.component.spec.ts
REQ-PROJ-03	FR-PROJ-03	4.1.1	UC-PROJ-01	project-list.component.spec.ts

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-PROJ-04	FR-PROJ-04	4.1.1	UC-PROJ-01	app.routes.spec.ts, nav-items.spec.ts
REQ-PROJ-05	FR-PROJ-05, FR-PROJ-06	4.1.1	UC-PROJ-01	ProjectTeamEndpointsTests, ModuleBasedPermissionEndpointsTests, project-list.component.spec.ts
REQ-PROJ-06	FR-PROJ-07	4.1.1	UC-PROJ-01	project-list.component.spec.ts
REQ-PROJ-07	FR-PROJ-08	4.1.1	UC-PROJ-02	project-list.component.spec.ts
REQ-PROJ-08	FR-PROJ-09, FR-PROJ-10, FR-PROJ-11	4.1.1	UC-PROJ-02	ProjectTeamEndpointsTests, UserAllocationServiceTests, ManagerAssignmentTests, project-list.component.spec.ts (FR-PROJ-10 tasks kept: no automated test)
REQ-PROJ-09	FR-PROJ-12	4.1.1	UC-PROJ-01	project-list.component.spec.ts
REQ-PROJ-10	FR-PROJ-14	4.1.2	UC-PROJ-03	ProjectRequestValidatorsTests, UsersEndpointsTests, project-form-dialog.component.spec.ts
REQ-PROJ-11	FR-PROJ-15	4.1.2	UC-PROJ-03, UC-PROJ-04	project-form-dialog.component.spec.ts (server message shown; no backend test of the duplicate check)
REQ-PROJ-12	FR-PROJ-16	4.1.2	UC-PROJ-04	ProjectRequestValidatorsTests, project-form-dialog.component.spec.ts
REQ-PROJ-13	FR-PROJ-17	4.1.2	UC-PROJ-03	project-form-dialog.component.spec.ts, project-list.component.spec.ts, project-team-dialog.component.spec.ts
REQ-PROJ-14	FR-PROJ-18	4.1.2	UC-PROJ-03, UC-PROJ-04	project-form-dialog.component.spec.ts
REQ-PROJ-15	FR-PROJ-19	4.1.3	UC-PROJ-05	ProjectTeamEndpointsTests, project-team-dialog.component.spec.ts
REQ-PROJ-16	FR-PROJ-20, FR-PROJ-21	4.1.3	UC-PROJ-05	employee-picker.component.spec.ts, project-team-dialog.component.spec.ts
REQ-PROJ-17	FR-PROJ-22	4.1.3	UC-PROJ-05	UserAllocationServiceTests, project-team-dialog.component.spec.ts
REQ-PROJ-18	FR-PROJ-23	4.1.3	UC-PROJ-05	UserAllocationServiceTests, employee-picker.component.spec.ts

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-PROJ-19	FR-PROJ-24, FR-PROJ-30	4.1.3	UC-PROJ-05	ProjectTeamEndpointsTests, UserAllocationServiceTests, project-team-dialog.component.spec.ts
REQ-PROJ-20	FR-PROJ-25	4.1.3	UC-PROJ-05	ProjectTeamEndpointsTests, UserAllocationServiceTests, project-team-dialog.component.spec.ts
REQ-PROJ-21	FR-PROJ-26	4.1.3	UC-PROJ-05	UserAllocationServiceTests
REQ-PROJ-22	FR-PROJ-21, FR-PROJ-27	4.1.3	UC-PROJ-05	UserAllocationServiceTests, ProjectTeamEndpointsTests, project-team-dialog.component.spec.ts
REQ-PROJ-23	FR-PROJ-28	4.1.3	UC-PROJ-05	ProjectTeamEndpointsTests, UserAllocationServiceTests
REQ-PROJ-24	FR-PROJ-29	4.1.3	UC-PROJ-05	ProjectTeamEndpointsTests, UserAllocationServiceTests, project-team-dialog.component.spec.ts
REQ-PROJ-25	FR-PROJ-31	4.1.3	UC-PROJ-05	project-team-dialog.component.spec.ts expects a new row to start with no position (conflicts with FR-PROJ-31)
REQ-PROJ-26	FR-PROJ-32	4.1.3	UC-PROJ-06	ProjectTeamEndpointsTests, UserAllocationServiceTests, position-combobox.component.spec.ts
REQ-PROJ-27	FR-PROJ-33	4.1.3	UC-PROJ-06	ProjectTeamEndpointsTests, position-combobox.component.spec.ts, project-team-dialog.component.spec.ts
REQ-PROJ-28	FR-PROJ-34	4.1.3	UC-PROJ-06	ProjectTeamEndpointsTests, UserAllocationServiceTests
REQ-PROJ-29	FR-PROJ-35	4.1.3	UC-PROJ-06	ProjectTeamEndpointsTests, UserAllocationServiceTests, UserProjectAllocationsMigrationTests
REQ-PROJ-30	FR-PROJ-36	4.1.3	UC-PROJ-06	ProjectTeamEndpointsTests, UserAllocationServiceTests (no permission from a position); Manual check (no rename / delete)
REQ-PROJ-31	FR-PROJ-37	4.1.4	UC-PROJ-07	project-list.component.spec.ts
REQ-PROJ-32	FR-PROJ-38	4.1.4	UC-PROJ-07	project-detail.component.spec.ts
REQ-PROJ-33	FR-PROJ-39, FR-PROJ-44	4.1.4	UC-PROJ-07	project-detail.component.spec.ts

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-PROJ-34	FR-PROJ-40, FR-PROJ-44	4.1.4	UC-PROJ-07	ProjectTaskTests, project-detail.component.spec.ts
REQ-PROJ-35	FR-PROJ-41	4.1.4	UC-PROJ-07	ProjectTaskTests, project-detail.component.spec.ts
REQ-PROJ-36	FR-PROJ-42	4.1.4	UC-PROJ-07	project-detail.component.spec.ts
REQ-PROJ-37	FR-PROJ-43	4.1.4	UC-PROJ-07	project-detail.component.spec.ts
BR-PROJ-01	FR-PROJ-45	4.1.4	UC-PROJ-07	Manual UI (no test of organisation-wide access)
BR-PROJ-02	FR-PROJ-14, FR-PROJ-15	4.1.2	UC-PROJ-03	ProjectRequestValidatorsTests, project-form-dialog.component.spec.ts
BR-PROJ-03	FR-PROJ-27	4.1.3	UC-PROJ-05	UserAllocationServiceTests, ProjectTeamEndpointsTests
BR-PROJ-04	FR-PROJ-22, FR-PROJ-23	4.1.3	UC-PROJ-05	UserAllocationServiceTests
BR-PROJ-05	FR-PROJ-24, FR-PROJ-25	4.1.3	UC-PROJ-05	ProjectTeamEndpointsTests, UserAllocationServiceTests
BR-PROJ-06	FR-PROJ-26	4.1.3	UC-PROJ-05	UserAllocationServiceTests
BR-PROJ-07	FR-PROJ-28	4.1.3	UC-PROJ-05	ProjectTeamEndpointsTests, UserAllocationServiceTests
BR-PROJ-08	FR-PROJ-32, FR-PROJ-33, FR-PROJ-34, FR-PROJ-36	4.1.3	UC-PROJ-06	ProjectTeamEndpointsTests, UserAllocationServiceTests, position-combobox.component.spec.ts
BR-PROJ-09	FR-PROJ-09, FR-PROJ-10	4.1.1	UC-PROJ-02	ProjectTeamEndpointsTests, UserAllocationServiceTests (FR-PROJ-10 tasks kept: no automated test)
BR-PROJ-10	FR-PROJ-30	4.1.3	UC-PROJ-05	ManagerAssignmentTests, UserAllocationServiceTests, ManagerAssignmentEndpointsTests
BR-PROJ-11	FR-PROJ-13, FR-PROJ-45	4.1.1, 4.1.4	UC-PROJ-01, UC-PROJ-07	Manual UI

Coverage: 37 of 37 `REQ-PROJ` requirements and 11 of 11 `BR-PROJ` rules trace to at least one FR; 0 untraced.

12. Approvals

Role	Name	Decision	Date
Product owner	Jojo George	Pending	–
Reviewer	Jojo George	Pending	–

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